

Joint-Reservoir Second-Order Thermal Response for MQM Capture

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Abstract—We formalize the second-order Gaussian displacement response for the MQM flat-band capture reference at unchanged inverse temperature $5/16$ and deadline 16 . Although the retained correction is an eight-dimensional system matrix, its interaction-picture operators evolve jointly with the vacuum reservoir. For every finite passive Gaussian bath regularization, the response equals a covariance-weighted one-photon-launch output minus its vacuum normalization term. This identity preserves noncommuting dynamics while requiring only total-excitation sectors one and two. An independent double-commutator control agrees with that formulation after disclosed quadrature refinement. We also derive finite-bath photon-number tail bounds, distinguishing them from Gaussian perturbation-order bounds. No continuum response enclosure, complete thermal target, native physical compiler, or hardware validation is claimed.

Index Terms—Quantum control, thermal response, bosonic reservoirs, Gaussian states, excitation sectors.

I. SOURCE, BASIS AND VERIFICATION GATE

We retain the MQM thermal reference [1] with flat band $[\Omega - B, \Omega + B]$ with $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$, and $\lambda^2 = 4\sqrt{2}$. The clock-auxiliary basis is ordered

$$(p0, m0, e0, d1, d0, p1, m1, e1), \quad (1)$$

where the digit denotes auxiliary occupation. The system charge is

$$N_S = \text{diag}(1, 1, 1, 1, 0, 2, 2, 2), \quad (2)$$

and $S = I_C \otimes \sigma_R^-$ lowers that charge by one. In the rotating frame the coherent port Hamiltonian is

$$H_C = \sum_{r=0}^1 [h|pr\rangle\langle mr| + g|mr\rangle\langle er| + \text{h.c.}] \quad (3)$$

$$+ \lambda(|e0\rangle\langle d1| + |d1\rangle\langle e0|). \quad (4)$$

The common carrier is removed mathematically using total excitation conservation; it is not an unrestricted physical energy shift.

For a finite M -mode reference,

$$H_0 = H_C + \sum_j \delta_j b_j^\dagger b_j + \sum_j c_j (S^\dagger b_j + S b_j^\dagger), \quad (5)$$

with $\omega_j = \Omega + \delta_j > 0$ and c_j fixed by the selected flat-band quadrature. The stationary free thermal bath has occupation covariance N , with convention $N_{jk} = \langle b_k^\dagger b_j \rangle$. It is diagonal in this star-frequency basis; a chain transformation generally makes it non-diagonal. The launch is $|p0\rangle$ and the computational bank/reference are arbitrary identity spectators.

The inherited continuum vacuum error ceiling is $E_0 = 3/2^{18} + 2^{-128}$. Prior Gaussian pairing [2] bounds all fourth- and higher mean response terms by 2^{-25} . The remaining sufficient criterion is

$$\frac{1}{2} \|\Delta_2\|_1 < \frac{127}{2^{25}} - 2^{-128}. \quad (6)$$

Here Δ_2 is the *fixed-launch retained-state correction*, not a completely positive channel by itself. The finite regularizations below are implementation controls. Their vacuum outputs cannot silently replace the certified continuum vacuum output.

II. JOINT NONCOMMUTING INTEGRAL

Let $U_0(s) = e^{-iH_0 s}$, $\rho_* = |p0, \text{vac}\rangle\langle p0, \text{vac}|$, and

$$S_s = U_0(s)^\dagger (S \otimes I) U_0(s). \quad (7)$$

The displaced drive in this interaction picture is $D_s = \zeta_s S_s^\dagger + \zeta_s^* S_s$. The field is circular Gaussian with $\mathbb{E}\zeta_s \zeta_u = 0$ and

$$C(s - u) = \mathbb{E}\zeta_s \zeta_u^* = \sum_j c_j^2 n_\beta(\omega_j) e^{-i\delta_j(s-u)}. \quad (8)$$

In the continuum the sum becomes the inherited flat-band covariance integral. Define the final retained map

$$\mathcal{R}_T(X) = \text{Tr}_B[U_0(T) X U_0(T)^\dagger]. \quad (9)$$

The exact second-order coefficient is

$$\Delta_2 = - \int_0^T ds \int_0^s du \mathcal{R}_T(\mathcal{B}_{su}), \quad (10)$$

$$\mathcal{B}_{su} = C(s - u)[S_s^\dagger, [S_u, \rho_*]] \quad (11)$$

$$+ C(u - s)[S_s, [S_u^\dagger, \rho_*]]. \quad (12)$$

The final $U_0(T)$ cannot be omitted before taking the bath trace. Neither can S_s be replaced by the bare eight-dimensional S ; H_0 includes exchange with the vacuum bath. The two covariance orientations distinguish upward and downward noise. Taking a signed scalar kernel integral discards these operators.

The resulting 8×8 matrix is Hermitian and traceless. Number covariance implies $[\Delta_2, N_S] = 0$, so its trace norm decomposes into charge blocks of dimensions 1, 4, 3. This simplification holds for the declared fixed launch and passive bath; it is not an arbitrary-input architecture theorem.

III. EXACT ONE-PHOTON RESPONSE IDENTITY

Proposition 1. For a finite regularization of the declared stationary passive Gaussian bath with covariance N , let

$$\Gamma_1(N) = \sum_{jk} N_{jk} |1_j\rangle\langle 1_k|. \quad (13)$$

Then

$$\Delta_2^M = \mathcal{R}_{T,2}(|p0\rangle\langle p0| \otimes \Gamma_1(N)) \quad (14)$$

$$- (\text{tr } N) \rho_0^M(T), \quad (15)$$

where $\mathcal{R}_{T,2}$ is the same retained map restricted to total-charge sector two, and ρ_0^M is the vacuum-launch output propagated in sector one.

Proof. Introduce a bookkeeping parameter a that scales the bath covariance to aN . This is not physical temperature tuning. In a covariance eigenbasis each thermal occupation distribution has vacuum probability $(1+an_j)^{-1}$ and geometric ratio $an_j/(1+an_j)$. Therefore its total vacuum probability has derivative $-\sum_j n_j$ at $a = 0$, its one-photon block has derivative $\Gamma_1(N)$, and every block with at least two photons has zero first derivative. Passive mode rotation preserves total photon number and transforms the one-photon block as N .

Propagation under H_0 is independent of a . Taking the derivative of the exact finite thermal output thus yields (14). Alternatively, coherent displacement rescales the Gaussian drive as $\sqrt{a}\zeta$. Its averaged Dyson series is a power series in a ; the coefficient of a is exactly (10). Gaussian absolute pairing bounds justify the finite-mode differentiation and equality. $\square$

This identity is an exact response derivative, not an approximation of the full thermal state by one photon. Higher orders are handled by the separate Gaussian remainder bound. For the continuum, $\text{tr } N$ need not be finite; the two terms in (14) must be combined in each regularization before taking a controlled limit. A finite one-photon formula alone supplies no continuum error certificate.

IV. EXCITATION SECTORS AND RESOURCE ACCOUNTING

The conserved total charge is $Q = N_S + \sum_j b_j^\dagger b_j$. The system charge multiplicities are $(1, 4, 3)$ for charges $(0, 1, 2)$. Hence

$$\dim \mathcal{H}_0 = 1, \quad (16)$$

$$\dim \mathcal{H}_1 = 4 + M, \quad (17)$$

$$\dim \mathcal{H}_2 = 3 + 4M + \binom{M+1}{2}, \quad (18)$$

$$\dim \mathcal{H}_3 = 3M + 4\binom{M+1}{2} + \binom{M+2}{3}. \quad (19)$$

Equation (14) requires sectors one and two. A direct displacement-integral implementation additionally visits sector zero; no sector above two is needed for the paired second-order coefficient. At $M = 768 = 3 \cdot 2^8$, sector one has $772 = 4 \cdot 193$ states and sector two has 298371 states; sectors zero through two total 299144. These are mathematical Fock-sector dimensions, not physical qubit counts. Sparse propagation and bath partial trace are required for practical scaling.

TABLE I
FINITE MIDPOINT CONTROLS, NOT CONTINUUM CERTIFICATES

M	$\dim \mathcal{H}_1$	$\dim \mathcal{H}_2$	$\frac{1}{2} \ \Delta_2^M\ _1$
2	6	14	2.205×10^{-8}
4	8	29	6.500×10^{-8}
8	12	71	7.789×10^{-8}
16	20	203	1.343×10^{-7}

A safe rotating-frame bound on the $Q \leq 2$ generator is

$$\|H_0\|_{Q \leq 2} \leq 2B + h + g + \lambda + 2\sqrt{2}G < 64, \quad (20)$$

where $G^2 = \sum c_j^2 = 128/\pi$ for the frozen midpoint grids. Use $\pi > 3$, $G < 7$, $\lambda < 5/2$, and $\sqrt{2} < 3/2$. This is a sector bound, not an unrestricted Fock-space or physical energy-budget certificate. The earlier vacuum one-excitation bound cannot be reused for the hot sector.

V. FINITE-BATH PHOTON TRUNCATION

For M passive thermal modes with occupation eigenvalues bounded by n_* , the total initial photon count K obeys

$$\mathbb{E}(1+x)^K = \prod_j (1 - n_j x)^{-1}. \quad (21)$$

The coefficient of x^r is $\mathbb{E}\binom{K}{r}$. Since $\binom{K}{r} \geq 1$ for $K \geq r$, positivity gives

$$\Pr(K \geq r) \leq \binom{M+r-1}{r} n_*^r. \quad (22)$$

The source frequencies satisfy $\beta\omega \geq 15$. The verified positive Taylor lower sum $e^{15} > 2^{21}$ gives $n_* < 1/(2^{21} - 1)$. At $M = 768$, initial cutoffs $K \leq 1$ and $K \leq 2$ have discarded-weight ceilings below 2^{-23} and 2^{-36} , respectively.

Normalize the projected initial thermal state. Its half-trace distance from the original finite thermal state equals the discarded weight. Unitary propagation and partial trace preserve this allowance as a retained upper bound at every time. A launch cutoff $K \leq r - 1$ requires exact total sectors $Q \leq r$; individual bath occupations must allow r , since the system can emit its initial excitation. An occupation cap of only $r - 1$ would incorrectly block allowed dynamics.

This global photon tail grows with M and does not produce a uniform infinite-continuum cutoff. A continuum thermal state may not be a trace-class density on the vacuum Fock representation. The reduced Gaussian continuum remains the declared reference assumption. Continuum bath-transfer errors, covariance projection errors, numerical enclosures, and photon truncation allowances must be certified separately. Gaussian perturbation order, initial photon count, and conserved charge are distinct typed objects.

VI. FROZEN CONTROLS AND REMAINING EVIDENCE

The grids have $\delta_j = -16 + (j + 1/2)32/M$ and $c_j^2 = 128/(\pi M)$. Binary64 diagonalization evaluates (14). Hermiticity, zero trace, and charge-block structure pass their declared 10^{-10} checks. For $M = 2$, independent triangular Gauss-Legendre integration of (10) initially failed at order 128,

with Frobenius discrepancy 1.447×10^{-9} against the frozen 10^{-10} tolerance. The original recipe and failure receipt remain preserved. Refinement to orders 256 and 512 gives discrepancies 8.135×10^{-20} and 2.791×10^{-19} without changing source parameters or tolerance. These are numerical implementation checks, not directed intervals or independent external review.

To close (6), a continuum response approximation must include numerical and continuum-transfer debits inside its second-order budget. The cold localized-launch truncation receipt does not certify a thermal one-photon launch, whose covariance occupies bath modes throughout the regularization. A failure of a sufficient bound does not prove that cooling or spectral engineering is necessary. The primary target remains open; physical promotion is zero.

REFERENCES

- [1] B. Mayes, "A Gaussian thermal reference and independent memory-kernel audit for MQM capture," MQM internal research draft, version 0.1, 2026.
- [2] B. Mayes, "A frequency-resolved Gaussian channel bound for MQM thermal capture," MQM internal research draft, version 0.1, 2026.